



Team Details

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FALCONS

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Problem Statement Addressed

DRIFT-SENSE: AI-POWERED NAVIGATION-ERROR RECOVERY FOR WAFER INSPECTION TOOLS

Context & Industrial Need

Modern semiconductor wafer inspection tools (e.g., Applied Materials systems) must revisit exact site coordinates thousands of times per day across hundreds of dies per wafer to ensure rigorous process control and defect metrology.

Cause of Navigation Errors

Mechanical motion stages accumulate position drift over time due to thermal expansion, ambient cleanroom vibrations, and mechanical slack—causing tool landing offsets of several pixels away from the intended target.

Core Technical Challenge

Dense, highly periodic die structures (DRAM memory arrays, FinFET gates) present hundreds of visually identical repeating features per frame. Classical template-matching algorithms break down due to spatial periodicity and ambiguity.

Objective & Significance

Develop an advanced AI/CV algorithm to accurately locate a 10x scaled Reference Image pattern within a larger Search Image with nanometer precision, breaking ties by selecting the match closest to the search center.



Idea Description



Treat drift recovery as a visual-search problem, not a recalibration problem — ask the network ‘where is this known reference inside the current frame?’ instead of re-homing the stage from scratch.

KEY CONCEPT & APPROACH

Core idea: given one high-zoom reference capture and the current low-zoom search frame, directly regress the pixel location of that reference pattern — a learned, trainable stand-in for classical normalized cross-correlation (NCC) template matching.

Fundamental approach: cast localization as similarity learning — embed both images with one shared-weight CNN, correlate the embeddings, and read the peak off with a differentiable soft-argmax so the whole pipeline trains end-to-end on pixel-error, not on hand-tuned correlation thresholds.

Why this framing: on periodic semiconductor layouts (DRAM/FinFET-style repeats), the failure mode isn’t ‘no match’ — it’s several equally-good matches. A learned similarity function can be trained to prefer the right one; a fixed correlation score cannot.

SOLUTION OVERVIEW

Pipeline: shared 4-layer CNN backbone (16→32→64→64 ch) embeds reference + search → depth-wise cross-correlation produces a similarity heatmap → soft-argmax converts it to a continuous (x, y) in one forward pass.

What makes it fit this problem: a dual loss (coordinate MSE + Gaussian-heatmap term, $\sigma=8\text{px}$) directly penalizes confidence on the wrong repeated peak — it targets the periodic-layout failure mode explicitly, not just average distance.

Delivery: a single 60,656-parameter model exported to ONNX (249.9 KB) with a standalone CLI script — sized to sit inside existing tool-control software as a drop-in recovery step, not a new compute stack.



Proposed Solution

SOLUTION DETAILS

DriftSense recovers stage position after navigation drift by finding a known reference die pattern, captured at 10× zoom, inside a wider low-zoom search frame — a lightweight Siamese cross-correlation network (SiamFC-style). A shared 4-layer CNN backbone (16→32→64→64 channels) encodes both images; the reference feature map is used as a depth-wise convolution kernel over the search feature map, producing a similarity heatmap that a differentiable soft-argmax head converts directly into (x, y) pixel coordinates — no discrete grid search, no hand-tuned correlation threshold. The model exports to ONNX (249.9 KB, opset 13) alongside a standalone inference.py, so it can run as an inline recovery step inside existing tool-control software rather than a separate compute stack.

DATASET PLAN & CLASS DESIGN

Reference-search pairs (current): 750 (525 train / 105 valid / 120 test), each annotated with bounding box (x1,y1,x2,y2), center (center_x, center_y) and pattern rotation (rotation_deg).

Scale relationship: reference captured at 10× the search image's zoom (SCALE_RATIO = 10.0) — search resized 1000→512 px, reference template resized to 51×51 px to preserve the physical scale ratio.

Loss design: combined L2 coordinate loss + Gaussian-heatmap cross-entropy ($\sigma = 8$ px) — explicitly suppresses probability mass on repeated 'look-alike' peaks in periodic DRAM/FinFET-style patterns.

Optimization: Adam, lr = 1e-3, cosine-annealing schedule, batch size 8, 15 epochs, best checkpoint selected by validation pixel error.

Evaluation protocol: mean / median / 90th-percentile pixel error plus success-rate @5/10/20/50 px on the held-out test split, broken out by |rotation| bucket to isolate rotation robustness.

Deployment artifacts: model_weights.pt (247.8 KB) + model.onnx (249.9 KB) + model_onnx_config.json, runnable via one CLI call (inference.py --reference ... --search ... --weights ...) for drop-in grading or tool integration.

Technology & Feasibility / Methodology Used



IMPLEMENTATION AND STRATEGY A custom SiCNN (Siamese Cross-Correlation Network) architecture is implemented for fast, sub-pixel SEM pattern localization. The lightweight model utilizes shared-weight feature extraction and cross-correlation matching to deliver high precision while minimizing computational overhead. The trained PyTorch model is exported to ONNX format, enabling low-latency, Python-free execution on edge inspection tools



Software Architecture

Search/Reference Input Preprocessing ->
Shared 4-Block CNN Feature Extraction -> 2D
Cross-Correlation (F.conv2d)-> Bilinear
Upsampling -> Continuous Soft-Argmax
Localization Head



Hardware Components Training:

- Cloud/Workstation GPU (e.g., NVIDIA T4) achieving full convergence in minutes (~15 epochs)
- **Inference:** Lightweight CPU systems execution (<1s latency)
- **Deployment:** Edge-ready for embedded metrology tool controllers via ONNX runtime



Development Tools

- PyTorch for deep learning model design and cross-correlation pipeline
- OpenCV / NumPy for spatial image scaling and preprocessing
- ONNX for cross-platform model serialization and edge deployment



BASELINE MODEL AND RESULTS

Parameter / Metric	Specification / Value
Architecture	SiCNN (Custom 4-Block Siamese CNN)
Matching Mechanism	Differentiable Cross-Correlation + Soft-Argmax
Framework	PyTorch
Export Format	ONNX (FP32)
Input Dimensions	Search: 512 x 512 px Template: 51 x 51 px
Model Footprint	~247 KB (~10 ⁵ parameters)
Inference Speed	< 1 second on standard CPU
Localization Precision	Sub-pixel accuracy (< 1 px target coordinate error)



Innovation and Uniqueness

A learned, differentiable alternative to classical template matching — built specifically for the repeated, look-alike structures of periodic layouts.



KEY INNOVATION

Depth-wise Siamese cross-correlation (SiamFC-style): a shared 4-layer CNN backbone encodes the reference patch and search frame; the reference feature map is used as a per-channel convolution kernel over the search feature map to produce a similarity heatmap.

A differentiable soft-argmax head turns that heatmap into a continuous (x,y) prediction, trained jointly with a Gaussian-weighted heatmap loss — this term explicitly penalizes probability mass on repeated 'look-alike' peaks, the core failure mode on periodic DRAM/FinFET-style layouts.

This combines an existing tracking architecture (SiamFC) with a custom dual-loss formulation for periodic-pattern disambiguation — not a new base algorithm.

COMPETITIVE ADVANTAGE

Classical normalized cross-correlation (NCC) is deterministic and needs no training, but scores every location independently — on periodic layouts, the true match and its look-alikes can score similarly, making the argmax pick ambiguous by construction.

Our network instead learns embeddings plus a loss that explicitly separates the true peak from neighbors, and outputs a sub-pixel-capable continuous coordinate rather than a discrete grid cell.

Footprint: 60,656 params / 249.9 KB ONNX — small enough for on-tool inference. Current accuracy (mean 287 px on our 120-pair test set) has not yet been benchmarked head-to-head vs. an NCC baseline; that comparison and accuracy tuning are the immediate next steps, not a claimed result today.



Impact and Benefits



DriftSense targets navigation-error recovery on inspection/metrology tools: re-localizing a known reference pattern inside a wider search frame after stage drift, instead of a full re-scan/recalibration cycle.

Primary Impact

Replaces full stage re-alignment / re-scan cycles with a single reference-vs-search inference pass, so a drifted tool can attempt local re-localization before falling back to a full recalibration routine.

Architecture is deployment-sized: 60,656 trainable parameters, exported to a 249.9 KB ONNX graph — light enough to run alongside existing tool-control software rather than requiring a separate compute stack.

Quantifiable Outcomes

Trained/validated on 750 reference-search pairs (525/105/120 train/valid/test split), 15 epochs.

Validation pixel error fell 306 → 272.6 px (−11%) over training — evidence the learned cross-correlation signal is trainable on this data.

Current test-set accuracy (mean 287 px error) is a baseline, not yet production-grade; tightening it via more epochs/data is scoped as next-step work, not a claimed result.



GitHub & Video Link



GitHub Repository

 https://github.com/Roshni-K6/Drift_Sense_AI-Powered-Navigation-Error-Recovery.git



Prototype / Simulation Video

 https://drive.google.com/file/d/1EsBORXQLjaT6hq72Nqd0d_Md71xIXMom/view?usp=sharing



Research and References

Research Background & Methodology



We formulate SEM pattern localization as a **reference-to-search matching problem**. A Siamese CNN extracts shared features from the reference and search images, while cross-correlation generates a spatial localization heatmap. The reference is resized to account for the **10× scale difference**. A soft-argmax converts the heatmap into the predicted **(x,y)** center. Training combines coordinate regression with a Gaussian heatmap objective to handle periodic DRAM/FinFET patterns and competing matches. Image-quality variations are introduced during training to improve robustness, and performance is measured using pixel localization error on a held-out test set.



References & Citations

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(PDF) [Image Matching by Normalized Cross-Correlation](#)